

LOCARNO-MONTI, Switzerland

WMO: 067600

Lat: 46.1722N

Lon: 8.7875E

Elev: 1247

StdP: 14.05

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.9	29.9	1.8	6.3	38.5	6.4	8.0	40.9	12.0	45.0	10.5	45.6	2.4	10	0.375

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.0	86.9	69.3	84.3	68.2	81.9	66.9	71.5	83.8	70.0	81.4	68.7	79.2	5.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.3	105.1	77.0	66.0	100.3	75.9	64.8	96.0	74.7	36.0	84.0	34.7	81.7	33.6	79.4	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.3	9.4	7.7	24.4	90.5	3.8	3.3	21.6	92.9	19.4	94.8	17.3	96.6	14.6	99.0		
(5)			21.2	73.8	3.2	1.4	18.9	74.8	17.0	75.6	15.2	76.3	12.8	77.3	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	55.9	39.5	41.9	49.4	55.2	62.0	69.2	72.8	71.6	64.3	56.3	46.9	40.6	(6)	(7)
	DBStd	12.89	4.95	5.73	6.51	6.32	5.80	5.74	4.76	4.72	4.91	5.20	5.24	5.15	(7)	(8)
	HDD50	1099	329	233	91	22	1	0	0	0	0	10	119	294	(8)	(9)
	HDD65	4024	791	646	485	297	127	25	5	6	69	272	544	757	(9)	(10)
	CDD50	3242	4	6	71	179	372	575	706	668	428	206	25	2	(10)	(11)
	CDD65	694	0	0	1	4	33	151	246	209	48	2	0	0	(11)	(12)
	CDH74	4168	0	0	5	21	161	942	1647	1220	164	8	0	0	(12)	(13)
	CDH80	981	0	0	0	3	18	224	423	300	12	1	0	0	(13)	(14)
(14) Wind	WSAvg	2.8	2.3	2.6	3.2	3.4	3.3	3.3	3.2	2.9	2.8	2.5	2.3	2.1	(14)	(15)
(15) Precipitation	PrecAvg	68.0	2.9	2.6	3.2	5.8	7.6	7.1	6.6	7.0	7.5	7.1	7.6	2.9	(15)	(16)
	PrecMax	100.7	9.6	10.9	10.9	15.7	21.8	17.6	12.8	12.6	17.4	22.7	29.2	8.9	(16)	(17)
	PrecMin	39.1	0.0	0.2	0.2	0.5	2.8	1.1	1.2	0.8	0.8	0.4	0.1	0.0	(17)	(18)
	PrecStd	14.9	2.4	2.4	2.5	3.8	4.1	3.7	3.1	2.9	5.1	4.8	6.9	2.3	(18)	(19)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.0	63.0	72.4	76.6	82.5	89.1	90.0	91.0	81.6	74.0	62.7	57.7	(19)	(20)
		MCWB	46.3	46.3	51.6	57.5	63.9	69.2	70.8	71.4	66.9	59.2	50.0	43.1	(20)	(21)
	2%	DB	53.0	56.3	66.3	72.2	78.2	85.3	86.9	86.4	78.3	69.2	58.7	52.6	(21)	(22)
		MCWB	41.7	43.7	49.6	55.5	61.5	68.5	69.5	69.3	64.7	58.6	48.8	42.1	(22)	(23)
	5%	DB	49.1	53.1	62.7	69.2	75.1	82.5	84.5	83.0	75.4	66.4	56.0	49.5	(23)	(24)
		MCWB	39.7	42.1	48.3	53.5	59.8	67.2	68.5	67.8	63.1	56.9	48.2	41.6	(24)	(25)
	10%	DB	46.2	50.2	59.5	65.9	72.0	79.4	82.0	80.4	72.6	63.7	53.7	47.0	(25)	(26)
		MCWB	38.8	40.7	46.8	52.1	58.4	65.6	67.2	66.6	61.5	56.3	47.9	41.0	(26)	(27)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.4	48.0	53.6	59.7	66.1	72.8	73.8	72.8	69.0	62.8	54.6	48.5	(27)	(28)
		MCDB	58.8	59.2	66.1	72.9	79.6	85.1	86.0	86.6	79.1	68.5	57.1	51.2	(28)	(29)
	2%	WB	43.9	45.3	51.5	57.2	63.7	70.3	71.7	71.0	66.8	61.0	52.5	45.7	(29)	(30)
		MCDB	49.7	53.2	62.4	69.6	75.3	82.2	84.0	83.4	75.3	65.7	55.3	48.3	(30)	(31)
	5%	WB	42.0	43.6	49.9	55.3	61.9	68.5	70.1	69.6	65.2	59.5	51.1	43.9	(31)	(32)
		MCDB	46.8	50.6	59.4	66.1	72.0	79.8	81.6	80.4	72.5	63.9	53.7	47.3	(32)	(33)
	10%	WB	40.3	42.1	48.4	53.7	60.2	67.0	68.8	68.3	63.7	58.0	49.5	42.2	(33)	(34)
		MCDB	44.6	48.2	56.7	62.6	69.1	77.4	79.5	77.9	70.3	62.2	52.1	46.0	(34)	(35)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.6	11.6	14.3	14.5	14.6	15.4	16.0	15.2	13.3	10.9	9.1	8.6	(35)	(36)
		MCDBR	14.3	15.8	18.9	19.4	19.2	18.7	18.7	18.5	16.7	14.6	12.5	12.0	(36)	(37)
		MCWBR	8.4	9.1	9.9	9.7	9.4	8.8	8.5	8.3	8.2	8.0	7.5	7.6	(37)	(38)
	5% WB	MCDBR	11.7	13.8	16.2	17.0	16.6	17.0	17.0	16.4	14.4	11.2	8.3	9.0	(38)	(39)
		MCWBR	7.6	8.6	9.1	8.9	8.6	8.3	8.2	7.8	7.5	6.6	5.8	6.3	(39)	(40)
(40) Clear-Sky Solar Irradiance	taub		0.307	0.341	0.400	0.465	0.478	0.495	0.480	0.470	0.447	0.411	0.348	0.303	(40)	(41)
	taud		2.299	2.218	2.108	1.992	2.037	2.053	2.108	2.144	2.166	2.216	2.325	2.347	(41)	(42)
	Ebn at Noon		247	259	258	251	252	248	251	247	241	230	228	235	(42)	(43)
	Edh at Noon		25	33	43	53	52	52	48	45	40	33	24	22	(43)	(44)
(44) All-Sky Solar Radiation	RadAvg		495	785	1183	1421	1595	1864	1876	1613	1240	791	484	412	(44)	(45)
	RadStd		64	98	164	208	149	145	146	128	109	103	76	55	(45)	(46)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.77	N/A	N/A	+1.51	N/A	N/A	N/A	N/A	+126	+126
(47) Regional (116 neighbors)		N/A	N/A	N/A	+1.30	+0.72	+0.54	N/A	N/A	+104	+29

Nomenclature: See separate page